$C^{(1)\,i1\_i2\_i3\_i4\_}_{qq} \to$

$$\frac{1}{16\pi^2}\Big(\frac{1}{2592}\frac{1}{m_\Phi^2}\Big(s_\gamma^2\,\overline{y_d}^{i4p}\big(y_d^{i3p}\big(162\,c_\gamma^2\,\overline{y_u}^{i2r}y_u^{i1r}+(5g_1^2+24g_3^2)\,\delta_{i1i2}\big)-36g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$s_\gamma^2\,\overline{y_d}^{i2p}\big(-36g_3^2\,y_d^{i3p}\,\delta_{i1i4}+$$
$$y_d^{i1p}\big(162\,c_\gamma^2\big(-\overline{y_d}^{i4r}y_d^{i3r}+\overline{y_u}^{i4r}y_u^{i3r}\big)+(5g_1^2+24g_3^2)\,\delta_{i3i4}\big)\big)+$$
$$c_\gamma^2\big(\overline{y_u}^{i4p}\big(y_u^{i3p}(-13g_1^2+24g_3^2)\,\delta_{i1i2}-36g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(-36g_3^2\,y_u^{i3p}\,\delta_{i1i4}+y_u^{i1p}\big(-162\,s_\gamma^2\,\overline{y_u}^{i4r}y_u^{i3r}+(-13g_1^2+24g_3^2)\,\delta_{i3i4}\big)\big)\big)\Big)+$$
$$\frac{1}{36}g_3^4(3\delta_{i1i4}\delta_{i2i3}-2\delta_{i1i2}\delta_{i3i4})\,\mathrm{LF}_{3,0}[m_3]+\frac{1}{24}g_3^4(3\delta_{i1i4}\delta_{i2i3}-2\delta_{i1i2}\delta_{i3i4})\,\mathrm{LF}_{4,-1}[m_3]+$$
$$\frac{2}{45}g_3^4(-3\delta_{i1i4}\delta_{i2i3}+2\delta_{i1i2}\delta_{i3i4})\,\mathrm{LF}_{5,-2}[m_3]+$$
$$\frac{1}{972}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,0}[m_{\tilde{d}}^{p}]-$$
$$\frac{5}{2592}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,-1}[m_{\tilde{d}}^{p}]+$$
$$\frac{1}{1215}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{5,-2}[m_{\tilde{d}}^{p}]+$$
$$\frac{1}{162}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\delta_{i3i4}-\frac{5}{432}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\delta_{i3i4}+\frac{1}{324}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{5}{864}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\delta_{i3i4}+\frac{1}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{972}\sum_p\big(54g_3^4\delta_{i1i4}\delta_{i2i3}+(g_1^4-36g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]-$$
$$\frac{5}{2592}\sum_p\big(54g_3^4\delta_{i1i4}\delta_{i2i3}+(g_1^4-36g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]+$$
$$\frac{1}{1215}\sum_p\big(54g_3^4\delta_{i1i4}\delta_{i2i3}+(g_1^4-36g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]+$$
$$\frac{1}{972}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(4g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,0}[m_{\tilde{u}}^{p}]-$$
$$\frac{5}{2592}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(4g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,-1}[m_{\tilde{u}}^{p}]+$$
$$\frac{1}{1215}\sum_p\big(27g_3^4\delta_{i1i4}\delta_{i2i3}+2(4g_1^4-9g_3^4)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{5,-2}[m_{\tilde{u}}^{p}]+$$
$$\frac{1}{216}\Big(s_\gamma^2\,\overline{y_d}^{i4p}\big(y_d^{i3p}\big(27c_\gamma^2\,\overline{y_u}^{i2r}y_u^{i1r}+2(g_1^2+3g_3^2)\delta_{i1i2}\big)-9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+s_\gamma^2\,\overline{y_d}^{i2p}$$
$$\big(-9g_3^2\,y_d^{i3p}\,\delta_{i1i4}+y_d^{i1p}\big(27c_\gamma^2\big(\overline{y_d}^{i4r}y_d^{i3r}+\overline{y_u}^{i4r}y_u^{i3r}\big)+2(g_1^2+3g_3^2)\delta_{i3i4}\big)\big)+$$
$$c_\gamma^2\big(\overline{y_u}^{i4p}\big(2y_u^{i3p}(-2g_1^2+3g_3^2)\delta_{i1i2}-9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(-9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+y_u^{i1p}\big(27s_\gamma^2\,\overline{y_u}^{i4r}y_u^{i3r}+2(-2g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\big)\Big)$$
$$\mathrm{LF}_{1,2}[m_\Phi]+\frac{1}{144}\Big(-s_\gamma^2\,\overline{y_d}^{i4p}y_d^{i3p}\big(9c_\gamma^2\,\overline{y_u}^{i2r}y_u^{i1r}+g_1^2\delta_{i1i2}\big)+$$
$$s_\gamma^2\,\overline{y_d}^{i2p}y_d^{i1p}\big(9s_\gamma^2\,\overline{y_d}^{i4r}y_d^{i3r}-9c_\gamma^2\,\overline{y_u}^{i4r}y_u^{i3r}-g_1^2\delta_{i3i4}\big)+$$
$$c_\gamma^2\big(g_1^2\,\overline{y_u}^{i4p}y_u^{i3p}\delta_{i1i2}+\overline{y_u}^{i2p}y_u^{i1p}\big(9c_\gamma^2\,\overline{y_u}^{i4r}y_u^{i3r}+g_1^2\delta_{i3i4}\big)\big)\Big)\,\mathrm{LF}_{2,1}[m_\Phi]+$$
$$\frac{1}{1296}g_1^2\Big(9\big(s_\gamma^2\,\overline{y_d}^{i4p}y_d^{i3p}-c_\gamma^2\,\overline{y_u}^{i4p}y_u^{i3p}\big)\delta_{i1i2}+$$
$$\big(9s_\gamma^2\,\overline{y_d}^{i2p}y_d^{i1p}-9c_\gamma^2\,\overline{y_u}^{i2p}y_u^{i1p}+4g_1^2\delta_{i1i2}\big)\delta_{i3i4}\Big)\,\mathrm{LF}_{3,0}[m_\Phi]-$$
$$\frac{5}{864}g_1^4\,\mathrm{LF}_{4,-1}[m_\Phi]\,\delta_{i1i2}\delta_{i3i4}+\frac{1}{405}g_1^4\,\mathrm{LF}_{5,-2}[m_\Phi]\,\delta_{i1i2}\delta_{i3i4}+\frac{1}{324}g_1^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\delta_{i3i4}-\frac{2}{405}g_1^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{2592}\big(-225g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(-8g_1^2+75g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{324}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}+g_3^2(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i1}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{2592}\big(-225g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(-8g_1^2+75g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{324}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}+g_3^2(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i2}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{2592}\big(-225g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(-8g_1^2+75g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{324}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}+g_3^2(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i3}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{2592}\big(-225g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(-8g_1^2+75g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{324}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}+g_3^2(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i4}]+$$
$$\frac{1}{216}\big(\overline{y_d}^{i4p}\big(2y_d^{i3p}(g_1^2+3g_3^2)\delta_{i1i2}-9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(-9g_3^2\,y_d^{i3p}\,\delta_{i1i4}+2y_d^{i1p}(g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{p},\tilde{\mu}]+$$
$$\frac{1}{432}\big(\overline{y_d}^{i4p}\big(-2y_d^{i3p}(g_1^2+3g_3^2)\delta_{i1i2}+9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(9g_3^2\,y_d^{i3p}\,\delta_{i1i4}-2y_d^{i1p}(g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,2,-1}[m_{\tilde{d}}^{p},\tilde{\mu}]+$$
$$\frac{1}{432}\big(\overline{y_d}^{i4p}\big(-2y_d^{i3p}(g_1^2+3g_3^2)\delta_{i1i2}+9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(9g_3^2\,y_d^{i3p}\,\delta_{i1i4}-2y_d^{i1p}(g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{p},\tilde{\mu}]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_1]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_1]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_2]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_2]+$$
$$\frac{1}{1296}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}-2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_3]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_3]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_1]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_1]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_2]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_2]+$$
$$\frac{1}{1296}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}-2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_3]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_3]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i3},m_1]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i3},m_1]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i3},m_2]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i3},m_2]+$$
$$\frac{1}{1296}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}-2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i3},m_3]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i3},m_3]+$$
$$\frac{1}{7776}g_1^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i4},m_1]+$$
$$\frac{1}{15552}g_1^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i4},m_1]+$$
$$\frac{1}{288}g_2^2\big(-9g_3^2\delta_{i1i4}\delta_{i2i3}-(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i4},m_2]+$$
$$\frac{1}{576}g_2^2\big(9g_3^2\delta_{i1i4}\delta_{i2i3}+(g_1^2-6g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i4},m_2]+$$
$$\frac{1}{1296}\big(9g_3^4\delta_{i1i4}\delta_{i2i3}-2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i4},m_3]+$$
$$\frac{1}{2592}\big(-9g_3^4\delta_{i1i4}\delta_{i2i3}+2g_3^2(4g_1^2+3g_3^2)\delta_{i1i2}\delta_{i3i4}\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i4},m_3]+$$
$$\frac{1}{216}\big(\overline{y_u}^{i4p}\big(2y_u^{i3p}(-2g_1^2+3g_3^2)\delta_{i1i2}-9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(-9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+2y_u^{i1p}(-2g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{p},\tilde{\mu}]+$$
$$\frac{1}{432}\big(\overline{y_u}^{i4p}\big(2y_u^{i3p}(2g_1^2-3g_3^2)\delta_{i1i2}+9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+2y_u^{i1p}(2g_1^2-3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,2,-1}[m_{\tilde{u}}^{p},\tilde{\mu}]+$$
$$\frac{1}{432}\big(\overline{y_u}^{i4p}\big(2y_u^{i3p}(2g_1^2-3g_3^2)\delta_{i1i2}+9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+2y_u^{i1p}(2g_1^2-3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{p},\tilde{\mu}]+$$
$$\frac{1}{432}\big(\overline{y_d}^{i4p}\big(-2y_d^{i3p}(g_1^2+3g_3^2)\delta_{i1i2}+9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(9g_3^2\,y_d^{i3p}\,\delta_{i1i4}-2y_d^{i1p}(g_1^2+3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,1,0}[\tilde{\mu},m_{\tilde{d}}^{p}]+$$
$$\frac{1}{432}\big(\overline{y_d}^{i4p}\big(y_d^{i3p}(-5g_1^2+12g_3^2)\delta_{i1i2}-18g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(-18g_3^2\,y_d^{i3p}\,\delta_{i1i4}+y_d^{i1p}(-5g_1^2+12g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_{\tilde{d}}^{p}]+$$
$$\frac{1}{432}\big(\overline{y_d}^{i4p}\big(y_d^{i3p}(g_1^2-6g_3^2)\delta_{i1i2}+9g_3^2\,y_d^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_d}^{i2p}\big(9g_3^2\,y_d^{i3p}\,\delta_{i1i4}+y_d^{i1p}(g_1^2-6g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_{\tilde{d}}^{p}]+$$
$$\frac{1}{432}\big(\overline{y_u}^{i4p}\big(2y_u^{i3p}(2g_1^2-3g_3^2)\delta_{i1i2}+9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+2y_u^{i1p}(2g_1^2-3g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{2,1,0}[\tilde{\mu},m_{\tilde{u}}^{p}]+$$
$$\frac{1}{432}\big(\overline{y_u}^{i4p}\big(y_u^{i3p}(g_1^2+12g_3^2)\delta_{i1i2}-18g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(-18g_3^2\,y_u^{i3p}\,\delta_{i1i4}+y_u^{i1p}(g_1^2+12g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_{\tilde{u}}^{p}]+$$
$$\frac{1}{432}\big(\overline{y_u}^{i4p}\big(y_u^{i3p}(g_1^2-6g_3^2)\delta_{i1i2}+9g_3^2\,y_u^{i1p}\,\delta_{i2i3}\big)+$$
$$\overline{y_u}^{i2p}\big(9g_3^2\,y_u^{i3p}\,\delta_{i1i4}+y_u^{i1p}(g_1^2-6g_3^2)\delta_{i3i4}\big)\big)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_{\tilde{u}}^{p}]+$$
$$\frac{1}{5184}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}+\frac{1}{2592}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]$$
$$\delta_{i1i2}\delta_{i3i4}+\frac{1}{5184}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{2592}g_1^4m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+\frac{3}{64}g_2^4\,\mathrm{LF}_{2,1,1,-1}[m_2,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{3}{32}g_2^4m_2^2\,\mathrm{LF}_{2,1,1,0}[m_2,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{3}{64}g_2^4\,\mathrm{LF}_{2,1,1,-1}[m_2,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+\frac{3}{32}g_2^4m_2^2\,\mathrm{LF}_{2,1,1,0}[m_2,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{7}{96}g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}-\frac{1}{24}g_3^4m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{1}{144}g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}+\frac{5}{36}g_3^4m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{144}g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+\frac{5}{36}g_3^4m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{7}{96}g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}-\frac{1}{24}g_3^4m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{1}{8}\tilde{\mu}^2\big(\overline{y_d}^{i4p}y_d^{i3p}\,\overline{y_u}^{i2r}y_u^{i1r}+\overline{y_d}^{i2p}y_d^{i1p}\,\overline{y_u}^{i4r}y_u^{i3r}\big)\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_{\tilde{d}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{1}{16}\overline{y_d}^{i2p}\,\overline{y_d}^{i4r}y_d^{i1p}y_d^{i3r}\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_{\tilde{d}}^{r},m_{\tilde{d}}^{p}]+$$
$$\frac{1}{16}\overline{y_u}^{i2p}\,\overline{y_u}^{i4r}y_u^{i1p}y_u^{i3r}\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_{\tilde{u}}^{r},m_{\tilde{u}}^{p}]+$$
$$\frac{1}{288}g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{1}{144}m_1m_3g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}-$$
$$\frac{1}{432}g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{216}m_1m_3g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i3}]\,\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{432}g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_3,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{216}m_1m_3g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_3,m_{\tilde{q}}^{i2},m_{\tilde{q}}^{i4}]\,\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{288}g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{1}{144}m_1m_3g_1^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_1,m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{3}{32}g_2^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_2,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{3}{16}m_2m_3g_2^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_2,m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{3}{32}g_2^2g_3^2\,\mathrm{LF}_{1,1,1,1,-1}[m_2,m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}+$$
$$\frac{3}{16}m_2m_3g_2^2g_3^2\,\mathrm{LF}_{1,1,1,1,0}[m_2,m_3,m_{\tilde{q}}^{i3},m_{\tilde{q}}^{i4}]\,\delta_{i1i4}\delta_{i2i3}\Big)$$